



UNIVERSITY OF IBADAN

Faculty of Technology

Department of Agricultural and Environmental Engineering

TAE 414

PRACTICAL TERM PAPER

ON

**PRODUCTION, CURING, AND COMPRESSIVE STRENGTH TESTING OF
CONCRETE USING A 1:1:2 MIX RATIO**

Submitted by:

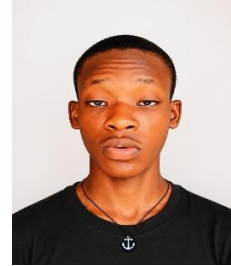
GROUP 4

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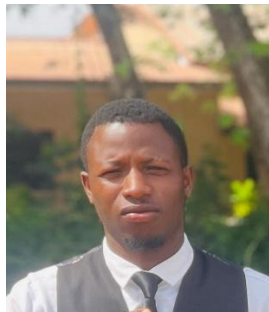
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ABSTRACT

Concrete is among the most widely employed construction materials in engineering, valued for its compressive strength, durability, and adaptability across a broad range of structural applications. This practical exercise involved the production of concrete specimens using a nominal 1:1:2 mix ratio of Ordinary Portland Cement, fine sand, and crushed granite. The constituent materials were batched by weight using a head pan and a weighing scale, blended with controlled amounts of water, cast into standard moulds, compacted to eliminate entrapped air, and subsequently cured in water. The prepared specimens were later subjected to compressive strength testing. Findings from the exercise underscored the critical roles of accurate material proportioning, thorough mixing, adequate compaction, and consistent water curing in governing the final strength and durability of the hardened concrete.